

The Linux File System Hierarchy

MODULE 1: GETTING STARTED & NAVIGATING THE SYSTEM

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Why is "Everything a File" in Linux?

- In Linux, the phrase "everything is a file" means that the operating system represents and accesses almost all system resources—including hardware devices, directories, and processes—through a uniform file-based interface.
- The Core Concept: Uniform Interface
- Linux treats hardware, processes, and storage as files so programs can interact with the entire system using the exact same basic actions: **Open, Read, Write,** and **Close**
- **Simplicity:** Developers only need to learn one consistent set of tools to talk to any resource.

Types of Files in Linux?

- In Linux, **Everything is treated as File.**

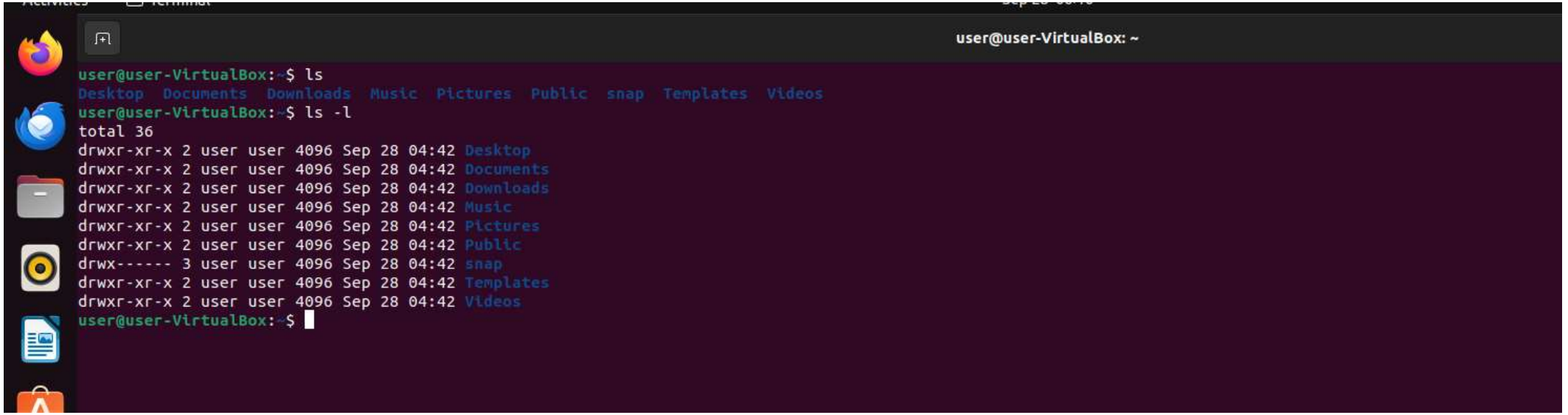
All files are divided into 3 types.

1. **Normal or Ordinary files:** A normal file can be either text files like (abc.txt, cde.txt) OR binary files like (images, videos etc)
2. **Directory files:** These files represent directories. : Windows uses the term **folder**, while Linux and Unix systems use the term **directory**, though both describe a container used to organize data.
3. **Device files:** In Linux, every device is represented as a file. By using this file we can communicate with that device.

How to check File Type

- In Ubuntu, blue color files represents directories and all remaining are considered as normal files. This color conventions are varied from flavor to flavor. It is not standard way to check file type.
- We have to use ‘ls –l’ command.
- The first character represents the type of file.
  - **d** -> Directory File
  - **-** -> Normal File
  - **l** -> Link File
  - **c**-> Character special File
  - **b** -> Block special File
  - **s** -> Socket File

Note : c,b,s are representing system files and mostly used by super user.



File System Navigation Commands

- **.** : represents current directory
- **..** : represents parent directory
- **$ cd .** : changes to current directory (useless)
- **$ cd ..** : changes to parent directory
- **$ cd** : if we are not passing any argument, then changes to user home directory.
- **$ cd ~** : changes to user home directory
- **$ cd –**: changes to previous working directory

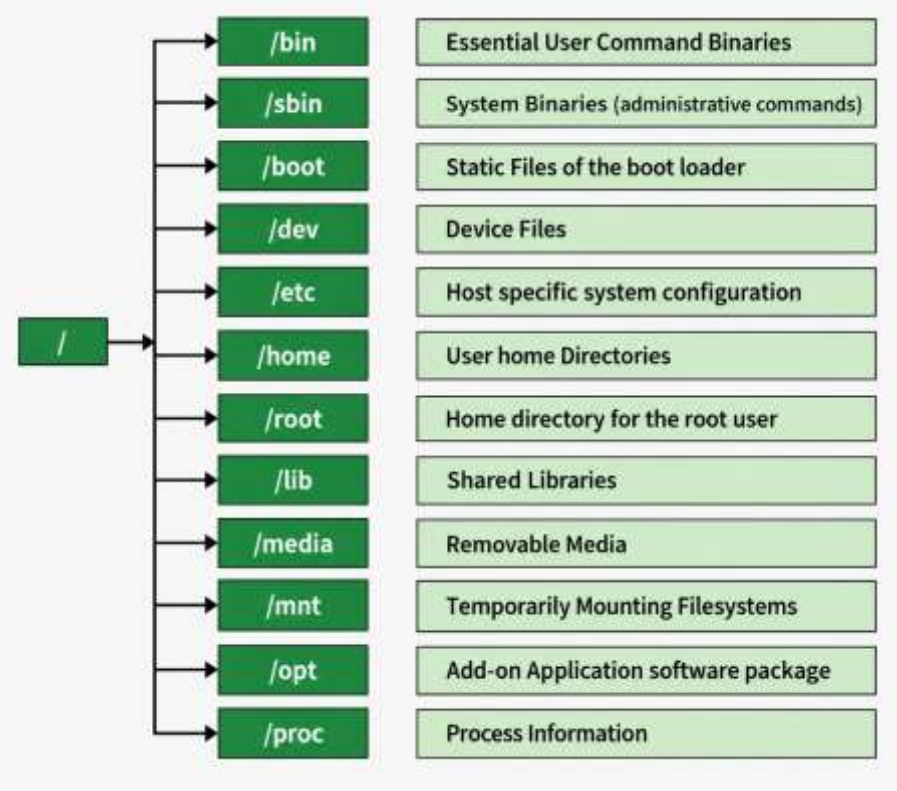
Linux File System Hierarchy

Directory	Name	Purpose
/	Root	The absolute base of the file system tree.
/bin	User Binaries	Contains essential command-line binaries (utilities) needed by all users (e.g., ls, cp, mv, rm)
/sbin	System Binaries	Contains crucial executable files used primarily by the system administrator (root) for system maintenance, booting, and administrative tasks.
/boot	Boot Loader	Stores static files required to boot the system.
/dev	Devices	Contains device nodes that represent system hardware (e.g., /dev/sda for a hard drive, /dev/null as a data sink)
/etc	Editable Text Configuration	The hub for host-specific, system-wide configuration files.

Directory	Name	Purpose
/home	Home Directories	Provides personal storage space for individual, non-root users (e.g., /home/username)
/root	Root Home	The dedicated personal home directory explicitly reserved for the administrative superuser (root).
/lib	System Libraries	Contains the essential shared libraries and kernel modules that the operating system needs to boot up and run basic commands.
/media	Removable Media	Serving as an automatic **mount point** for removable devices like USB drives, external hard disks, or CD-ROMs.
/mnt	Mount	A location reserved for system administrators to **manually mount** temporary filesystems or network shares.
/opt	Optional	Third-party, proprietary, or standalone applications that are not part of the core operating system and are typically not managed by the default distribution package manager (like apt or dnf).
/proc	Processes	A virtual, in-memory filesystem that tracks **real-time kernel information** and state values for active processes.
/sys	System	Another virtual filesystem providing specialized interface for **hardware devices and drivers**.

Directory	Name	Purpose
/run	Runtime Data	Contains **system runtime information.**
/tmp	Temporary	A public scratchpad for apps to dump volatile **temporary files,** often completely wiped automatically upon system reboot.
/usr	User System Resources	The largest directory segment; contains user applications, secondary binaries (/usr/bin), documentation, and source code.
/var	Variable Data	Stores dynamic, frequently changing files such as system logs (/var/log), print queues, and web caches.

Linux File Hierarchy Structure




```
user@ubuntu:~$ tree -L 1 /
/
├── bin
├── sbin
├── boot
├── dev
├── etc
├── home
├── root
├── lib
├── media
├── mnt
├── opt
└── proc

12 directories, 0 files
user@ubuntu:~$
```

Basic Linux Commands

- Linux commands are the primary way to interact with a Linux operating system through the terminal.
- Linux commands are organized by purpose:
  1. **Navigation & Directory Management**
  2. **File Operations**
  3. **Searching & Text Processing**
  4. **Permissions & System Basics**
- Learning basic Linux commands helps beginners understand how Linux works and use it efficiently for daily tasks.

Navigation & Directory Management

- These commands help you move around the system and organize your folders:
  - **pwd** – print working directory
  - **ls** – lists all files and folders in your current directory
  - **ls –l** – for detailed information
  - **ls –a** – to view hidden files
  - **mkdir** – creates a brand-new directory
  - **rmdir** – removes an empty directory

File Operations

- Use these commands to create, view, copy, and delete files.
  - **touch** – creates a blank, empty file.
  - **cat** – concatenates and displays the full contents of a file in the terminal
  - **head** – Views very beginning of a long file.
  - **tail** – Views the end of a long file.
  - **cp** – copies files or directories from one place to another.
  - **mv** – moves or renames files and folders.
  - **rm** – deletes files permanently.
  - **rm –r** – to delete a folder and everything inside it.

Searching & Text Processing

- These commands let you look through text or find specific data quickly
  - **grep** – Searches for specific words or patterns within text or files.
  - **find** - Searches for files and directories across your entire system layout based on filters like name or date.
  - **clear** - Clears the terminal screen to give you a clean slate

Permissions & System Basics

- Essential commands for managing permissions and seeing how the system is performing:
  - **sudo** – Executes a command with administrative (root) privileges.
  - **chmod** – Changes the file read, write, and execute permissions.
  - **df –h** - Shows how much available disk space you have in a human-readable format.
  - **top** or **htop** – Displays running processes and system resources in real-time.